

Material Safety Data Sheet

BOILER CARE 1604



1. Product and company identification

Trade name of product : BOILER CARE 1604
Product use : Water-boiler treatment.

Product dilution information : Not applicable

Supplier's information : Ecolab Water Care Division
5105 Tomken Road
Mississauga ON L4W 2X5
1-800-352-5326

Code : 922088

Date of issue : 12-November-2013

EMERGENCY HEALTH INFORMATION: 1-800-328-0026
Outside United States and Canada CALL 1-651-222-5352 (in USA)

2. Hazards identification

Product AS SOLD

Physical state : Liquid.
Emergency overview : DANGER !

CAUSES DIGESTIVE TRACT, EYE AND SKIN BURNS.
CAUSES RESPIRATORY TRACT IRRITATION.
Do not ingest. Do not get in eyes, on skin or on clothing. Avoid breathing vapours, spray or mists. Keep container closed. Use only with adequate ventilation. Wash thoroughly after handling.

Routes of entry : Skin contact, Eye contact, Inhalation, Ingestion

Potential acute health effects

Product AS SOLD

Eyes : Corrosive to eyes.
Skin : Corrosive to the skin.
Inhalation : Severely irritating to the respiratory system.
Ingestion : Causes burns to mouth, throat and stomach.

See toxicological information (Section 11)

Product AT USE DILUTION

Product is sold ready to use.

Product AT USE DILUTION

Product is sold ready to use.

3. Composition/information on ingredients

Product AS SOLD

<u>Name</u>	<u>CAS number</u>	<u>% by weight</u>
POTASSIUM HYDROXIDE	1310-58-3	7 - 13
SODIUM SULFITE	7757-83-7	1 - 5

Product AT USE DILUTION

<u>Name</u>	<u>CAS number</u>	<u>% by weight</u>
Product is sold ready to use.		

4. First-aid measures

Product AS SOLD		Product AT USE DILUTION
Eye contact	: In case of contact, immediately flush eyes with cool running water. Remove contact lenses and continue flushing with plenty of water for at least 15 minutes. Get medical attention immediately.	Product is sold ready to use.
Skin contact	: In case of contact, immediately flush skin with plenty of water for at least 15 minutes while removing contaminated clothing and shoes. Get medical attention immediately. Wash clothing before reuse. Clean shoes thoroughly before reuse.	
Inhalation	: If inhaled, remove to fresh air. If exposed person is not breathing, give artificial respiration or oxygen applied by trained personnel. Get medical attention.	
Ingestion	: If material has been swallowed and the exposed person is conscious, give small quantities of water to drink. Do not induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.	

5. Fire-fighting measures

Product AS SOLD

Auto-ignition temperature	: Not available.
Flammable limits	: Not available.
Hazardous thermal decomposition products	: Decomposition products may include the following materials: carbon dioxide carbon monoxide sulfur oxides metal oxide/oxides
Special remarks on fire hazards	: Not available.
Special remarks on explosion hazards	: Not available.
Fire-fighting media and instructions	: Use an extinguishing agent suitable for the surrounding fire. Dyke area of fire to prevent runoff. In a fire or if heated, a pressure increase will occur and the container may burst.
Special protective equipment for fire-fighters	: Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

6. Accidental release measures

Product AS SOLD		Product AT USE DILUTION
Personal precautions	: Initiate company's spill response procedures immediately. Keep people out of area. Put on appropriate personal protective equipment (see section 8). Do not touch or walk through spilt material.	Product is sold ready to use.
Environmental precautions	: Avoid contact of spilt material and runoff with soil and surface waterways.	
Methods for cleaning up	: Follow company's spill procedures. Keep people away from spill. Put on appropriate personal protective equipment (see section 8). Absorb/neutralise liquid material. Use a tool to scoop up solid or absorbed material and put into appropriate labelled container. Use a water rinse for final clean-up.	

7. Handling and storage

Handling	Product AS SOLD
	: Do not ingest. Do not get in eyes, on skin, or on clothing. Do not breathe vapour or spray. Keep container closed. Use only with adequate ventilation. Wash thoroughly after handling.
Storage	Product AS SOLD
	: Keep out of reach of children. Store in a closed container. Keep container tightly closed. Store between the following temperatures: 0 and 35°C

Product AT USE DILUTION

Product is sold ready to use.

8. Exposure controls/personal protection

Engineering measures	Product AS SOLD
	: Use only with adequate ventilation. If user operations generate dust, fumes, gas, vapour or mist, use process enclosures, local exhaust ventilation or other engineering controls to keep worker exposure to airborne contaminants below any recommended or statutory limits. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.
Personal protection	
Eyes	: Use chemical splash goggles. For continued or severe exposure wear a face shield over the goggles.
Hands	: Use chemical-resistant, impervious gloves.
Skin	: Use synthetic apron, other protective equipment as necessary to prevent skin contact.
Respiratory	: Wear appropriate respirator when ventilation is inadequate and occupational exposure limits are exceeded.

Product AT USE DILUTION

Product is sold ready to use.

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Occupational exposure limits		TWA (8 hours)			STEL (15 mins)			Ceiling			Notations
Ingredient	List name	ppm	mg/m ³	Other	ppm	mg/m ³	Other	ppm	mg/m ³	Other	
potassium hydroxide	US ACGIH 3/2012	-	-	-	-	-	-	-	2	-	[3]
	AB 4/2009	-	-	-	-	-	-	-	2	-	
	BC 4/2012	-	-	-	-	-	-	-	2	-	
	ON 7/2010	-	-	-	-	-	-	-	2	-	
	QC 9/2011	-	-	-	-	2	-	-	-	-	

[3]Skin sensitisation

9. Physical and chemical properties

Physical state	Product AS SOLD
	: Liquid.
Flash point	: > 100°C
Colour	: Brown [Dark]
Odour	: Faint odour
pH	: 13 [Conc. (% w/w): 100%]
Boiling/condensation point	: Not available.
Melting/freezing point	: Not available.
Relative density	: 1.11
Vapour pressure	: Not available.

Product AT USE DILUTION

Product is sold ready to use.

9. Physical and chemical properties

Vapour density : Not available.
Odour threshold : Not available.
Evaporation rate : Not available.
Viscosity : Kinematic (room temperature): 0 cm²/s (0 cSt)
Dispersibility properties : Not available.
Solubility : Not available.
LogK_{ow} : Not available.

10. Stability and reactivity

Product AS SOLD

Stability : The product is stable. Under normal conditions of storage and use, hazardous polymerisation will not occur.
Conditions of instability : Not available.
Reactivity : Highly reactive or incompatible with the following materials: acids.
 Reactive or incompatible with the following materials: organic materials and metals.
 Slightly reactive or incompatible with the following materials: alkalis.
Hazardous decomposition products : Under normal conditions of storage and use, hazardous decomposition products should not be produced.
Hazardous polymerisation : Under normal conditions of storage and use, hazardous polymerisation will not occur.

11. Toxicological information

Potential acute health effects

Product AS SOLD

Eyes : Corrosive to eyes.
Skin : Corrosive to the skin.
Inhalation : Severely irritating to the respiratory system.
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Product AS SOLD

Potential chronic health effects

Carcinogenic effects : No known significant effects or critical hazards.
Mutagenic effects : No known significant effects or critical hazards.
Teratogenic effects : No known significant effects or critical hazards.
Reproductive effects : No known significant effects or critical hazards.
Sensitization to Product : No known significant effects or critical hazards.
Synergistic products (toxicologically) : Not available.

Toxicity data

Product/ingredient name	Test	Route	Result	Species
sodium sulfite (2:1)	LD50	Dermal	>2000 mg/kg	Rat
	LD50	Oral	>2150 mg/kg	Rat
	LC50	Inhalation	>5.5 mg/l	Rat

Other adverse effects

Target organs : Contains material which may cause damage to the following organs: upper respiratory tract.

12. Ecological information

Ecotoxicity : Not reported

13. Disposal considerations

Product AS SOLD

Waste disposal : Avoid disposal. Attempt to use product completely in accordance with intended use. Disposal should be in accordance with applicable regional, national and local laws and regulations.

Product AT USE DILUTION

Product is sold ready to use.

14. Transport information

Certain shipping modes or package sizes may have exceptions from the transport regulations. The classification provided may not reflect those exceptions and may not apply to all shipping modes or package sizes.

Product AS SOLD

TDG

TDG Classification UN1814
 TDG Proper shipping name POTASSIUM HYDROXIDE, SOLUTION
 Class 8
 Packing group II

IMO/IMDG

IMO/IMDG Classification UN1814
 IMO/IMDG Proper shipping name POTASSIUM HYDROXIDE SOLUTION
 Class 8
 Packing group II

For transport in bulk, see shipping documents for specific transportation information.

15. Regulatory information

Product AS SOLD

WHMIS : Class E: Corrosive material

Product AT USE DILUTION

Product is sold ready to use.

Canada inventory : All components are listed or exempted.

This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations and the MSDS contains all the information required by the Controlled Products Regulations.

16. Other information

Product AS SOLD

Hazardous Material Information System (U.S.A.) :

Health	*	3
Flammability		0
Physical hazards		0

National Fire Protection Association (U.S.A.) :

16. Other information



Date of issue : 12-November-2013

Responsible name : Regulatory Affairs
1-800-352-5326

▣ Indicates information that has changed from previously issued version.

Notice to reader

The above information is believed to be correct with respect to the formula used to manufacture the product in the country of origin. As data, standards, and regulations change, and conditions of use and handling are beyond our control, NO WARRANTY, EXPRESS OR IMPLIED, IS MADE AS TO THE COMPLETENESS OR CONTINUING ACCURACY OF THIS INFORMATION.